

Banking Data Engineering Platform Project Documentation

Enterprise Banking ETL Platform using Databricks Free Edition, PySpark, Delta Lake and Medallion Architecture

1. Executive Summary

This project demonstrates an end-to-end banking data engineering platform using Databricks Free Edition. It implements ingestion, validation, Medallion Architecture, fraud detection, audit logging, SQL analytics and dashboards.

Key Points

- Modular design
- Enterprise practices
- Databricks Free Edition compatible
- GitHub ready

2. Business Problem

Banking transactions arrive from ATM, UPI, NEFT, RTGS, CARD, INTERNET BANKING, MOBILE BANKING and BRANCH channels. The platform automates ingestion and reporting.

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3. Technology Stack

Databricks Free Edition, PySpark, Delta Lake, SQL, Unity Catalog, Volumes, Python, GitHub.

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4. Architecture

Configuration, utilities and logging are separated from ETL notebooks. Bronze stores raw data, Silver validated data and Gold reporting data.

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5. Notebook Guide

99_Config, 98_Utilities, 97_Logger, Project Setup, Precheck, Master Load, Ingestion, Validation, Bronze to Silver, Fraud Detection, Gold Layer, SQL Analytics, Dashboard and Master Pipeline.

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6. Folder Structure

Document notebooks, sample_data, docs, architecture, sql and tests folders.

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7. Reference Data

Customers, Accounts, Branches, Banks, Currencies, Account Types and Transaction Types.

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8. Pipeline Flow

Reference Data -> Master Load -> Bronze -> Validation -> Silver -> Fraud -> Gold -> SQL Analytics
-> Dashboard.

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9. Validation

Mandatory columns, duplicate detection, amount limits, transaction type validation and rejection handling.

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10. Fraud Detection

High-value transactions and configurable business rules identify suspicious records.

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11. Gold Layer

Daily, branch and customer summaries are generated for analytics.

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12. SQL Analytics

Views and reusable SQL queries support reporting.

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13. Dashboard

Gold tables feed KPI cards, trends, branch performance and fraud reports.

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14. GitHub Setup

Maintain README, diagrams, notebooks, sample data, SQL and documentation.

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15. Interview Questions

Explain Medallion Architecture, Delta Lake, audit logging, validation rules, fraud detection, orchestration, optimization and incremental loading.

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16. Future Enhancements

Metadata-driven framework, incremental processing, watermarking, CDC, SCD Type 2 and CI/CD.

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17. Conclusion

The project demonstrates a complete ETL lifecycle suitable for learning and portfolio purposes.

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